

Control bicluster regulatory architecture: TF motif landscape, cCRE classes, and genic context

Integrating TF motif enrichment, ENCODE cCRE class annotations, and genic context reveals a clear left–right continuum across control biclusters, ranging from enhancer-like/intronic states to promoter-centric architecture. On the left (clusters A–C), cCRE compositions are enriched for CA–H3K4me3 and CA–CTCF subclasses with few promoter-like sites (PLS), and the genic context is dominated by intronic/other regions (Figures 2–3). Concordantly, the motif heatmap shows generally weak enrichment in the leftmost clusters, punctuated by a limited set of factors with left-biased signals (Figure 1). Through the center (D–H), clusters shift toward enhancer identities: dELS predominate with increasing pELS in F–G (Figure 2). Their genic distribution is mixed—substantial intron and exon alongside promoter (Figure 3)—and the motif landscape strengthens broadly, forming a mid-zone band of elevated enrichment (Figure 1). At the right extreme (I–O), PLS overwhelmingly dominate with minimal CA subclasses and only residual pELS (Figure 2), and regions localize predominantly to promoters near transcription start sites (TSSs; Figure 3). In parallel, motif enrichment becomes uniformly high across many TFs (Figure 1), consistent with dense TFBS occupancy at promoter-proximal elements.

Together, these patterns define a graded promoter–enhancer balance: early, intronic/CA-rich biclusters exhibit sparse or factor-specific signals; mid, enhancer-dominated groups display rising and broader TFBS enrichment; and late, PLS-dominated clusters consolidate near promoters with the strongest and most pervasive motif signals. The coordinated shift

across Figures 1–3 indicates a continuous transition from distal/intragenic enhancer-like states to promoter-centric regulatory architecture in controls.

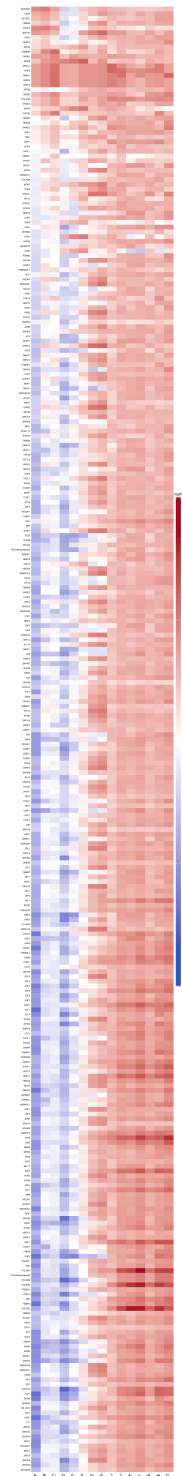


Figure 1. TFBS motif enrichment across control-region biclusters. *Heatmap of transcription factor binding motif enrichment across clusters of regulatory regions from control samples. Most motifs show lower enrichment in leftmost clusters and higher enrichment in rightmost clusters, with a smaller subset exhibiting the opposite pattern.*

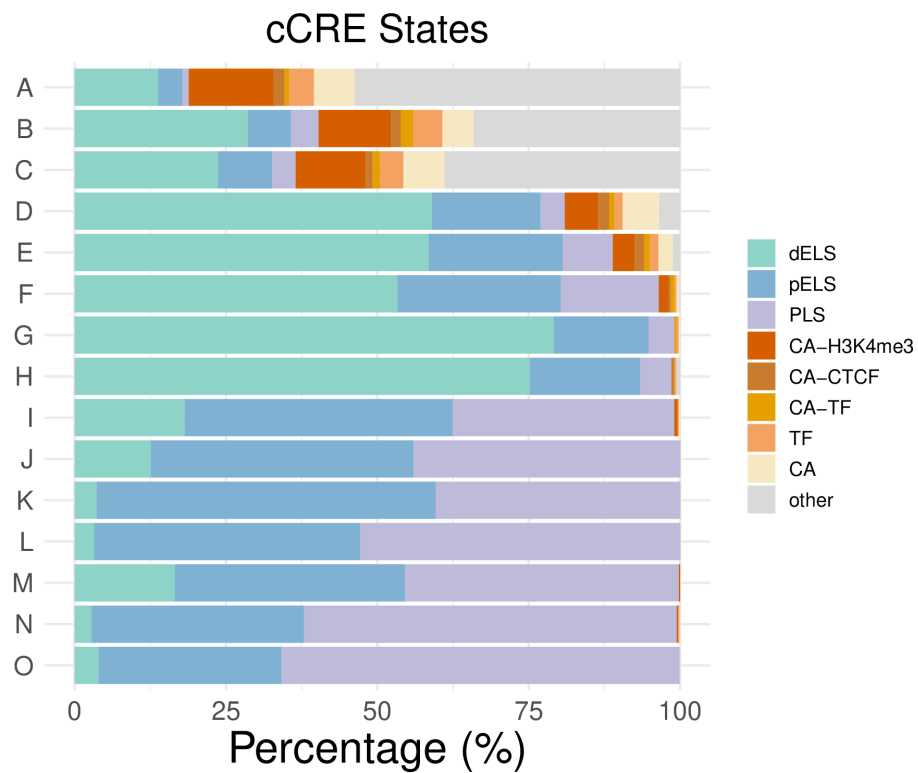


Figure 2. cCRE composition across control biclusters highlights promoter-enhancer balance. Proportions of cCRE states are summarized for bicluster groups A–O from the control analysis. Early groups emphasize enhancer-like and CA-associated categories, mid groups are mostly distal enhancers, and later groups are dominated by promoter-like (PLS) elements.

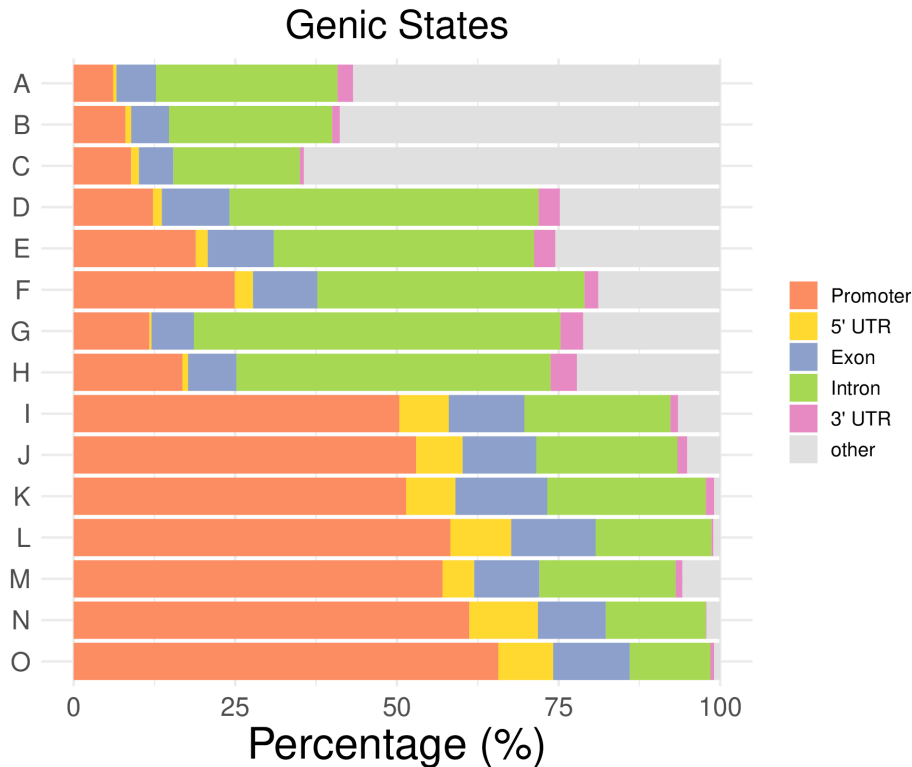


Figure 3. Genic location composition across C-group biclusters. Biclustered regions in the control group are summarized by their genomic feature assignments (promoter, 5' UTR, exon, intron, 3' UTR, other) for clusters A–O. Later clusters (H–O) are enriched at promoters with little other, whereas earlier clusters (A–C, G) are intron-rich and include more other; D–F show mixed genic annotations.

Treatment–control differential activity by cluster and pathway interpretation

Differential testing stratified by biclustering-defined region clusters shows that treatment-associated activity is concentrated in a subset of clusters and is directionally biased toward gains (Figure 4). No significant differences were detected for Cluster 1 or Cluster 3. Cluster 2 contained the largest number of differential regions in both directions but with a net excess of gains, whereas Cluster 4 exhibited a smaller overall response yet was similarly gain-skewed.

Pathway analysis of genes linked to significant regions revealed a sharp contrast in functional coherence between clusters (Figure 5). Despite its higher event count, Cluster 2 showed little to no enrichment across tested gene sets. In contrast, Cluster 4 displayed strong, selective enrichment led by `ESTROGEN_RESPONSE_EARLY` and `ESTROGEN_RESPONSE_LATE`, with additional prominent signals for `IL2_STAT5_SIGNALING` and `TNFA_SIGNALING_VIA_NFKB`; most other pathways, including metabolic and stress-response sets, were weak or not selectively enriched. This

divergence indicates that the smaller set of differential regions in Cluster 4 maps to a more unified program, whereas Cluster 2 harbors a broader but less coherent set of changes.

The dominance of estrogen-response hallmarks in Cluster 4 is consistent with activation of steroid hormone–regulated enhancer programs, wherein nuclear receptors and cofactors modulate chromatin binding and accessibility (Hiltunen et al., 2025). Concurrent enrichment of IL2/STAT5 and TNF α /NF- κ B modules suggests coordinated engagement of immune and inflammatory signaling that often intersects with steroid receptor–driven regulatory circuits at enhancers (Hiltunen et al., 2025). Together, these results support a model in which treatment preferentially augments regulatory activity in Cluster 4, yielding coherent pathway activation centered on estrogen-responsive and cytokine/NF- κ B

signaling, while the more numerous changes in Cluster 2 lack comparable pathway convergence.

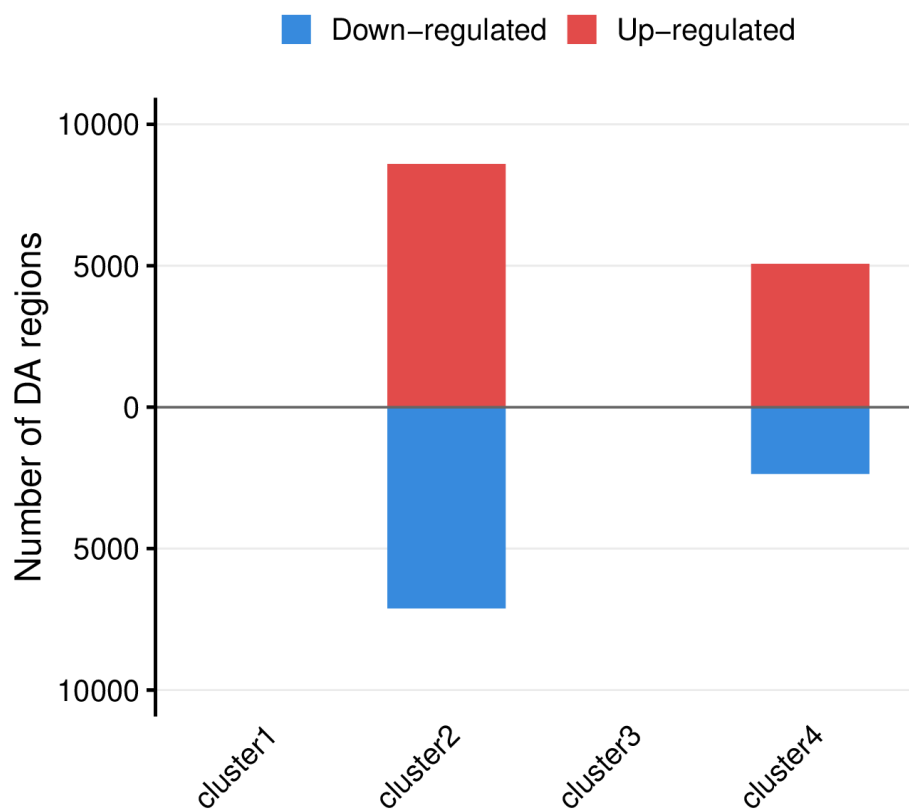


Figure 4. Differential signals by cluster favor up-regulation. *Counts of significant DA regions between treated and control are summarized per pair cluster. Differences are concentrated in clusters 2 and 4, each showing more up- than down-regulated regions, while clusters 1 and 3 show none.*

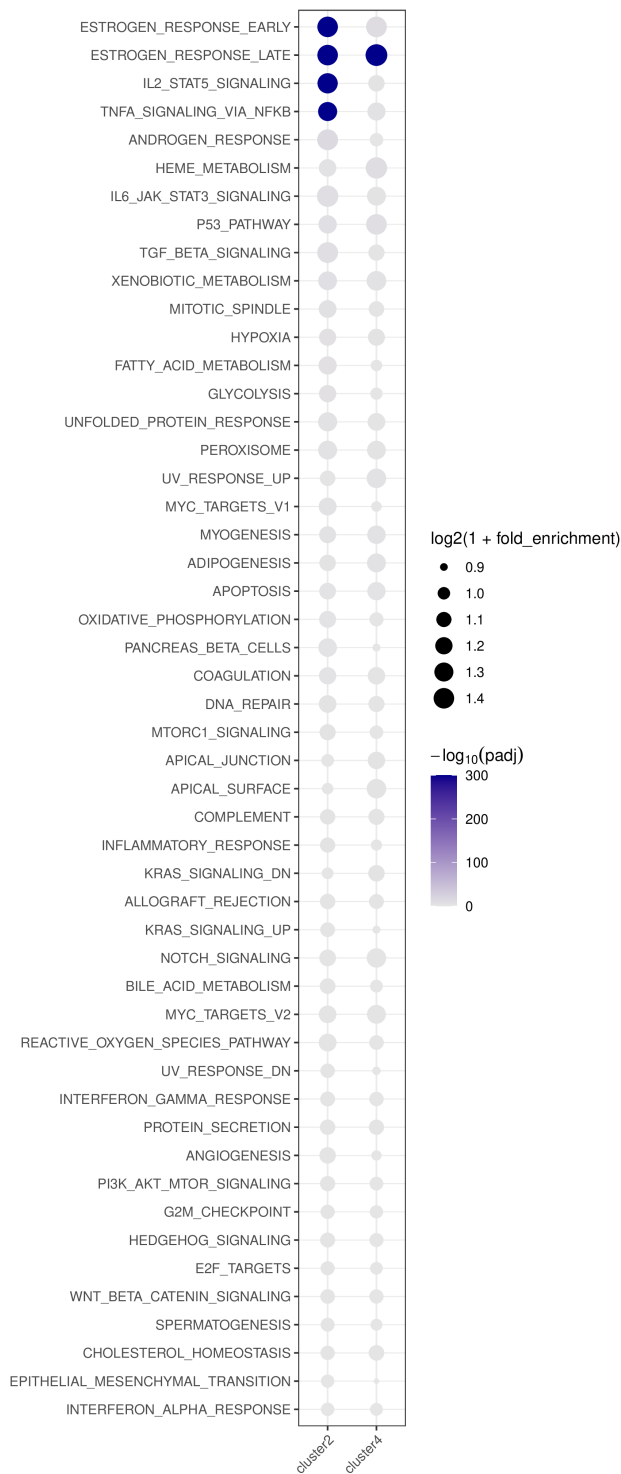


Figure 5. Selected pathways enriched among differential regions by cluster. Bubble size reflects fold enrichment and color indicates adjusted significance. Signals are concentrated in cluster4, with prominent enrichment of estrogen response (early and late), IL2–STAT5 signaling, and TNFA via NFKB, whereas most pathways show little enrichment in cluster2.

References

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